

# Amal Panwar

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• Birmingham, UK • [panwar.amal1995@gmail.com](mailto:panwar.amal1995@gmail.com) • +44-7849758548 • [linkedin.com/in/amal-panwar-031981119](https://www.linkedin.com/in/amal-panwar-031981119) • <https://github.com/amalpanwar>

## Skills & Interests

**Programming:** Python (NumPy, matplotlib, pandas, Scikit-learn), SQL, R (Shiny, Statistical Packages, ggplot2, rvest), Java, C/C++, DAX.

**Skills:** Mathematics (statistics and probability), Data Visualisation, Data mining, Microsoft Azure, Agile methodology, VSTS, SSIS/SSRS/SSAS, AI, Deep Learning, YOLO, openCV, RAG, LangChain, LLM, Time series analysis, Statistical analysis (Hypothesis testing, p-value), Problem-solving and Risk Management, Digital Transformation.

**Domain:** Insurance, Retail and Consumer Goods, Financial, Health, Sports.

**Tools and Technology:** Microsoft Office Suite (Dynamic 365, Power BI, Excel, etc.), CRM, Tableau, SQL Server Management Studio, Jira, Guidewire, NetLogo, Jupyter, Rstudio, Visual Studio Enterprise, Teradata, Jupyter, and Visio.

**Interests:** Football, Cricket, Tennis, Sports Analytics, General Knowledge, Political Science, Stock Market.

## Professional Experience

### Elgin City FC

Elgin, United Kingdom-Remote

#### Data Analyst/Scientist -Sports

June 2024 – Present

- Leveraged **WyScout** data and applied machine learning algorithms to identify high-potential player replacements, aiming to significantly enhance team performance.
- Conducted match analysis using **Computer Vision** techniques to support post-match evaluations, driving targeted improvements for future games.
- Created visual comparisons of target players against league benchmarks using **gauge charts** and **radar plots**, streamlining the scouting process and improving efficiency.
- Integrated data from diverse scouting reports to produce visuals that highlight player strengths and weaknesses, contributing to an anticipated **20%** improvement in team decision-making.
- Developed an **AI-powered** chatbot using the **RAG** pipeline, enabling scouts to interact dynamically with dashboards for enhanced usability and insights.

### Caribbean Goods

Glasgow, United Kingdom

#### Data/Statistical Analyst

May 2023 – Dec 2023

- Researched environmental factors affecting coffee sustainability in Brazil, Guatemala, Honduras, and Nicaragua using data mining techniques (KDD) to identify hidden patterns and actionable insights.
- Analyzed macroeconomic indicators to assess their influence on Coffee Futures prices.
- Achieved a mean absolute percentage error of less than 7% in price predictions using statistical and econometric modelling.
- Leveraged digital transformation and asset management strategies to help farmers optimize resource allocation, resulting in a 25% increase in sustainable coffee growth in tropical regions.

### Capgemini

Bengaluru, India

#### Data Analytics - Consultant

March 2022 – Aug 2022

- Extract and transform data in **Power BI** to fulfil a range of requests, creating insightful reports and identifying KPIs that align with business goals.
- Utilised the **Pytest** automation framework to validate **ETL** test cases and reduce **30%** testing time.

### Cognizant

Bengaluru, India

#### Junior Data Engineer/Analyst

Nov 2017 – March 2022

- Spearheaded development and implementation of a robust data warehouse framework, refining end-to-end data extraction, transformation, and loading (**ETL**) processes.

- Leveraged **PySpark** and **Azure Databricks** for scalable big data processing, reducing data pipeline execution time by **35%** and enhancing analytics capabilities.
- Utilised **T-SQL**, **MS Excel**, and **Power BI** to uncover valuable insights, detect patterns and identify trends within complex datasets.
- Executed Data Quality Analysis, identifying anomalies to enhance overall data quality by **20% - 30%**.
- Utilized machine learning modelling techniques in **Python** to develop a forecasting model, leading to a **15%** reduction in inventory write-offs.
- Thrived in collaborative team environments, demonstrating a proactive approach, strong multitasking abilities, and a history of autonomous contributions that significantly elevated data analysis and reporting processes.

## Relevant Projects

### University of Strathclyde

Glasgow, United Kingdom

#### Analysis of NHS Bed Availability

Nov 2022 – Nov 2022

- Conducted a comprehensive analysis of bed availability in Scotland's National Health Service (**NHS**) system.
- Proposed a data-driven methodology for anticipating future needs and enhancing bed allocation strategies by **30%**.
- Utilised **machine learning**-based forecasting models to predict inpatient bed demand, contributing to more efficient resource allocation.

#### [Football Analytics](#)

Jan 2024 – May 2024

- Engineered football goal prediction model using **R**.
- Boosted accuracy by **10-12%** through advanced statistical methods.
- Provided actionable insights using **Tableau** for strategic decision-making.
- Developed an **RShiny app** utilizing unsupervised machine learning to identify potential player signings.

## Education

### University of Strathclyde

Glasgow, UK

*Master of Science in Data Analytics.*

Sep 2022 – Sep 2023

- Award: Distinction
- Relevant coursework: Machine Learning, Data Science and Analytics in R, Big Data, Stochastic Modelling and Financial Econometrics.

### Galgotias University

Greater Noida, India

*Btech. Computer Science and Engineering.*

Sep 2013 – Sep 2017

- GPA: 8.02/10.
- Relevant coursework: Computer programming and problem-solving, Object-oriented programming, Data structure and algorithm, Database basics, Software development, Data mining and warehousing, Artificial Intelligence.

### References available on request:

- Liam Stewart (Elgin City FC, UK)
- Javier Gutierrez (Caribbean Goods, UK)
- Sandeep M (Cognizant, US)